

Advanced AI and Uncertainty

CS405 · Presentation 1 · Executive summary

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Purpose

Real data is noisy, vague and incomplete, so an AI that cannot say “I am not sure” will eventually be confidently wrong. The talk runs through eight parts and a live medical demo, comparing crisp logic, fuzzy logic and probability.

The eight parts in one line each

1	The AI landscape today	Symbolic and data-driven AI; search, knowledge, planning, reasoning.
2	Why uncertainty matters	Noisy sensors and vague language breed confident-but-wrong systems.
3	Kinds of imperfect information	Randomness, vagueness, imprecision, missing data: all in one record.
4	Aleatoric vs epistemic	Noise in the world stays; doubt in the model shrinks with more data.
5	Crisp logic and its limits	Hard lines flip between 29.9 and 30.0 °C; fine for rules like legal age.
6	Fuzzy logic: the basic idea	Degrees of membership (Zadeh, 1965); a degree is not a probability.
7	Probability and missing data	A positive test can mean 8%, not 90%; missing data has three mechanisms.
8	Live demo and the course	Pima diabetes demo in Colab; next: fuzzy inference, Bayesian networks.

What the demo showed (Pima Indians Diabetes data, 768 patients)

374 of 768

insulin values are hidden zeros (48.7%). A zero here means “not measured”, so it has to be treated as missing, not as a real reading.

0.48 vs 0.52

BMI 29.9 and 30.1 are almost the same patient. A crisp rule at 30 gives 0 vs 1; a fuzzy membership gives 0.48 vs 0.52.

0.68, 0.22, 0.07

spread of the model's prediction with $n = 20, 200$ and $2,000$ samples. Model doubt shrinks while the noise (about 3) stays.

Three things to take away

- 1 Crisp rules jump, fuzzy sets give degrees, and a degree of membership is not a probability.
- 2 More data shrinks the doubt in the model (epistemic) but never the noise in the world (aleatoric).
- 3 Missing data has mechanisms (MCAR, MAR, MNAR) and every fix has a price; a model you can trust is one that can say “I am not sure” and tell you why.

Demo notebook (Google Colab, view only)

https://colab.research.google.com/github/ahmedtayel714/Advanced-AI-and-Uncertainty--W1/blob/main/W01_UncertaintyDemo_Team01_MyBuild.ipynb

Data: Pima Indians Diabetes Database (NIDDK; Smith et al., 1988), 768 patients. Scan the code to open the notebook.

